Campus Companion

Feasibility Study

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1 Project idea

Campus Companion is a mobile application for students and visitors. Its main goal is to put useful campus information in one place and make buildings easier to find.

The application will work on Android and iOS. Its main feature is a small 3D map of our university campus. A user can search for a building, select it on the map and view a walking route. The application will also show announcements, events and campus services.

We chose this project because new students often lose time searching for rooms, offices and services. Information is also divided between websites, social-media pages and posters.

2 Project scope

2.1 Features for the first version

- sign-in with a test or university account;
- announcements and events;
- a list of buildings and campus services;
- search by building name or category;
- a simple 3D map of the campus;
- selection of buildings and points of interest;
- walking routes between important places;
- a small web page for adding or updating content.

2.2 Feature left for later development

QR code check-in is left for a later development phase. In that version, each student will have a QR code that can be scanned by an authorized staff or administrator application to record entry or attendance.

3 Technology choice

Part	Technology	Why we chose it
Mobile app	React Native and TypeScript	One project can run on Android and iOS.
3D map	Three.js and Blender	Blender is used to create the campus, and Three.js displays it.
Backend	Java and Spring Boot	The team can build a clear REST API in Java.
Database	PostgreSQL	It is suitable for users, buildings, events and routes.
Admin page	React or Next.js	It gives a simple way to update campus information.

The 3D map will use simple low-poly models instead of realistic buildings. This keeps the work manageable and helps the map run on normal phones.

4 Simple architecture

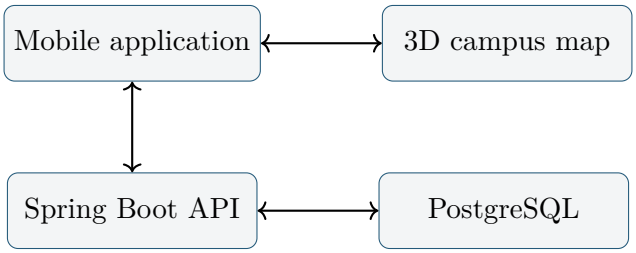


Figure 1: Basic project architecture.

The phone displays the screens and the 3D map. The Java backend manages users and campus content. PostgreSQL stores the information. The 3D models are created in Blender and loaded by the map.

5 Feasibility

5.1 Technical feasibility

The project is technically possible with the selected tools. React Native handles the normal application screens, and Three.js handles the 3D map. Spring Boot and PostgreSQL are enough for the backend of a student project.

The most difficult part will be creating the campus model. We will start with a small area containing a few important buildings. If it works well on our test phones, we can add the rest of the campus.

5.2 Time feasibility

The project can be completed during one semester if the team keeps the first version small.

Work	Time	Result
Planning and design	2 weeks	Screens, database and task division
Backend and database	4 weeks	Login and campus information API
Mobile screens	4 weeks	Main pages, search and events
3D map and routes	4 weeks	Campus model and basic navigation
Testing and presentation	2 weeks	Bug fixes and final demonstration

5.3 Cost feasibility

The main software is free and open source. We can develop and test locally using our own computers and phones. A small online demonstration may use free or student hosting. The main cost is the time needed to create the 3D buildings.

6 Main risks

Risk	Our response
The 3D map is slow	Use simple models, small textures and fewer buildings.
The project becomes too large	Keep only the features listed for the first version.
Campus information is incomplete	Begin with the information that the university approves.
The team falls behind schedule	Divide the work early and demonstrate a small working version first.

7 Conclusion

Campus Companion is feasible as a student project. The selected technologies are accessible to the team and do not require expensive tools. The project should begin with a prototype containing a few buildings, search and one walking route. If that prototype works correctly, the team can complete the other screens and extend the map.